Queen Mary
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# EBU5376
# Digital Signal Processing

Week 4 - Tutorial

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# Tutorial Outline

- Finite Impulse Response (FIR) filter
  - Window functions
- Infinite Impulse Response (IIR) filter
  - Bilinear transform
  - Frequency warping
- Cut-off frequency

# Tutorial Question 1

- Finite Impulse Response (FIR) filters can be constructed using the window method. There are three commonly used windows: Rectangular window, Hamming window, Hanning window. The equations of the windows are given as follows:

$$w_{rec}[n] = 1, \qquad -M \leq n \leq M$$

$$w_{ham}[n] = 0.54 + 0.46\cos\left(\frac{n\pi}{M}\right), \qquad -M \leq n \leq M$$

$$w_{han}[n] = 0.5 + 0.5\cos\left(\frac{n\pi}{M}\right), \qquad -M \leq n \leq M$$

Given a filter with an impulse response of $h[n] = [0.50, 0.55, 0.70, 0.55, 0.50], n = -2, \ldots 2$. Apply the Hamming window and the Hanning window to the filter, respectively.

# Tutorial Solution 1

The window function is applied by multiplying it with the impulse response

$$h_w[n] = h[n] \cdot w[n]$$

For Hamming window, the window coefficients are

$$w_{ham}[0] = 1, \qquad w_{ham}[-1] = w_{ham}[1] = 0.54, \qquad w_{ham}[-2] = w_{ham}[2] = 0.08$$

Then, multiply the window coefficients and the impulse response element-wise:

$$h_{ham}[0] = 0.7, \qquad h_{ham}[-1] = h_{ham}[1] = 0.2970, \qquad h_{ham}[-2] = h_{ham}[2] = 0.04$$

For Hanning window, the window coefficients are

$$w_{han}[0] = 1, \qquad w_{han}[-1] = w_{han}[1] = 0.5, \qquad w_{han}[-2] = w_{han}[2] = 0$$

Hence, the windowed impulse response is:

$$h_{han}[0] = 0.7, \qquad h_{han}[-1] = h_{han}[1] = 0.275, \qquad h_{han}[-2] = h_{han}[2] = 0$$

# Tutorial Question 2

Given an analog filter with a transfer function of $H_a(s) = \frac{1}{s+1}$, you are going to convert it to a digital filter using a sampling rate of 2 Hz.

i. The Bilinear Transform is a method to obtain the discrete-time approximation of a continuous-time system. State the equation of Bilinear Transform and describe the meaning of each parameter in the equation.

ii. Using Bilinear Transform, obtain the transfer function of the digital filter.

# Tutorial Solution 2

i. The equation for Bilinear Transform is

$$s = \frac{2}{T}\frac{1 - z^{-1}}{1 + z^{-1}}$$

where $s$ is a continuous parameter, $z$ is a discrete parameter, $T$ is the sampling interval

More particularly, $s$ and $z$ are the operators of Laplace transform and z-transform, respectively.

ii. The sampling rate is 2Hz, hence $T = 0.5$ seconds

The digital transfer function is obtained through

$$H(z) = H(s)\Big|_{s=\frac{2}{T}\frac{1-z^{-1}}{1+z^{-1}}} = \frac{1}{4\dfrac{1 - z^{-1}}{1 + z^{-1}} + 1} = \frac{1 + z^{-1}}{5 - 3z^{-1}} = \frac{0.2 + 0.2z^{-1}}{1 - 0.6z^{-1}}$$

# Tutorial Question 3

This question is about Butterworth filters.

i. An analog Butterworth filter is defined with an amplitude response of

$$|H_a(\omega)| = \frac{1}{\sqrt{1 + \left(\frac{\omega}{\omega_c}\right)^{2n}}}$$

where $\omega$ denotes the frequency, $n$ denotes the order, and $\omega_c$ denotes the cut-off frequency. Explain what is meant by the 'order' of a Butterworth filter and how it relates to the filter's transfer function.

ii. Write down the analog transfer function of an order-1 Butterworth filter with a cut-off frequency of $\omega_c = 1$ rad/s.

iii. Using Bilinear Transform, construct a digital filter using the Butterworth prototype defined in part ii. Assume the sampling rate is 4Hz. You don't need to pre-warp the frequency.

# Tutorial Solution 3

i. The 'order' of a Butterworth filter refers to the power of the polynomial in its transfer function. It determines the number of poles in the filter's transfer function and, consequently, the steepness or roll-off rate of the filter's magnitude response: the higher the order, the faster the attenuation as the frequency moves away from the cutoff frequency.

ii. $H(s) = \frac{1}{s+1}$

iii. The sampling rate is 4Hz, hence $T = 0.25$ seconds
The digital transfer function is obtained through

$$H(z) = H(s)\Big|_{s=\frac{2}{T}\frac{1-z^{-1}}{1+z^{-1}}} = \frac{1}{8\frac{1-z^{-1}}{1+z^{-1}}+1} = \frac{1+z^{-1}}{9-7z^{-1}} = \frac{0.11+0.11z^{-1}}{1-0.78z^{-1}}.$$

# Tutorial Question 4

This question asks you to create a digital filter using a Butterworth prototype.

i. Describe the characteristic of the passband response of a Butterworth filter

ii. By using Bilinear Transform, the relationship between the analog frequency $\Omega$ and the digital frequency $\omega$ is $\Omega = \frac{2}{T}\tan\frac{\omega T}{2}$, where $T$ is the sampling interval. Based on this relationship, describe the disadvantage of Bilinear Transform.

iii. We want to construct a digital filter with a cut-off frequency of 2 rad/s. Derive the cut-off frequency of the analog prototype, assuming the sampling rate is 2Hz.

iv. Using Bilinear Transform, construct a digital filter using the following Butterworth prototype with cut-off frequency denoted by $\omega_c$:

$$H(s) = \frac{1}{\frac{s}{\omega_c} + 1}$$

Assume the sampling rate is 2Hz. The resulting digital filter must have a cut-off frequency of 2 rad/s. Derive the transfer function of the digital filter.

# Tutorial Solution 4

i. Butterworth filters have flat pass-band response.

ii. Frequency Warping: The bilinear transform introduces non-linearity in the frequency scale, which results in frequency warping. High frequencies especially can be significantly distorted, and careful prewarping must be done if specific frequency characteristics need to be preserved.

iii. Substitute $\omega = 2$ rad/s and $T = \frac{1}{2}$ sec into $\Omega = \frac{2}{T}\tan\frac{\omega T}{2}$, we get

$$\Omega = 4\tan\frac{1}{2} \approx 2.18$$

Hence, the analog cut-off frequency is 2.18 rad/s

# Tutorial Solution 4

iv. The analog prototype is

$$H(s) = \frac{1}{\frac{s}{2.18} + 1} = \frac{2.18}{s + 2.18}$$

Perform Bilinear Transform :

$$\Rightarrow H(z) = H(s)\Big|_{s=\frac{2}{T}\frac{1-z^{-1}}{1+z^{-1}}}$$

In this question, $T = 1/2$, hence

$$H(z) = \frac{2.18}{4\frac{1-z^{-1}}{1+z^{-1}} + 2.18} = \frac{2.18 + 2.18z^{-1}}{4 - 4z^{-1} + 2.18 + 2.18z^{-1}} = \frac{2.18 + 2.18z^{-1}}{6.18 - 1.82z^{-1}}$$

$$= \frac{0.3528 + 0.3528z^{-1}}{1 - 0.2945}$$

# Tutorial Question 5

Figure 1 shows the amplitude response (on linear scale) of a digital filter. Based on this plot, answer the following questions:

i. Describe the filter represented by this amplitude response.

ii. Estimate the cut-off (normalized) frequency from this plot.

iii. Suppose the sampling rate is 4Hz, what is the estimated cut-off frequency in Hz?

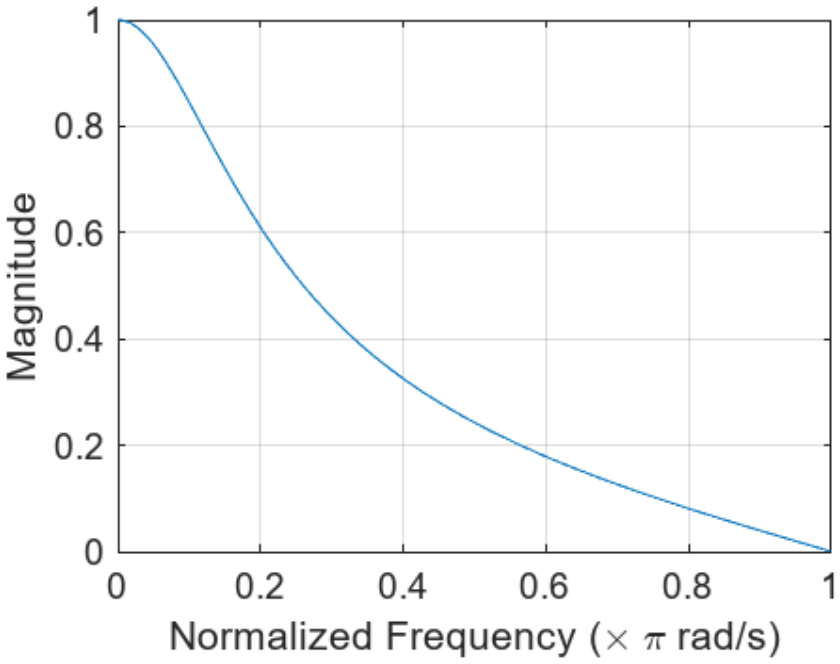


Figure 1

# Tutorial Solution 5

i. Low-pass filter. Because
   - high magnitude at low frequencies
   - a gradual decline in magnitude as frequency increases.
   - the filter allows low frequencies to pass while attenuating higher frequencies

ii. Identify the -3 dB point, which is 0.707 on a linear magnitude. This point on the plot appears to be the normalized frequency of around $0.15\pi$ rad/sample.

iii. normalized frequency = $\pi$*actual frequency / Nyquist frequency
Here, Nyquist frequency = 4/2=2Hz
For a normalized cut-off frequency of $0.15\pi$, the actual cut-off frequency is $0.15 \times 2 = 0.3$ Hz.

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